

Recursive Green's Function Method for Graphene

A Review and Application to Ballistic Quantum Transport

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August–November 2024

Project Type: Term Project
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Abstract

The recursive Green's function (RGF) method is a powerful numerical technique for calculating electronic transport in mesoscopic systems. It avoids direct inversion of the full open-system Hamiltonian by dividing a device into slices and recursively constructing the Green's functions associated with left-connected, right-connected, and fully connected subsystems.

This project reviews the Landauer and Landauer–Büttiker formalisms, Green's functions, scattering matrices, lead self-energies, Fisher–Lee relations, spectral functions, and local density of states. The method is then applied to single-layer graphene in the ballistic regime using a nearest-neighbor tight-binding model. Different slicing strategies for armchair and zigzag graphene geometries are discussed, with emphasis on reducing the number of sites per slice and inter-slice couplings. Conductance and local density of states are identified as key observables obtainable from the recursive construction.

The study demonstrates that RGF is especially useful for larger mesoscopic systems and provides a structured route from microscopic lattice Hamiltonians to experimentally relevant transport quantities.

1 Motivation

Quantum transport through nanoscale conductors is governed by wave interference, contact properties, confinement, and the availability of propagating modes. Direct diagonalization or inversion of the full device-plus-leads Hamiltonian becomes increasingly expensive as the device size grows.

The RGF method exploits the quasi-one-dimensional structure of a device. By dividing the conductor into slices, the full Green's function can be assembled recursively from smaller matrix blocks. This reduces memory and computational requirements while retaining the effects of semi-infinite leads through self-energy terms.

Graphene provides a useful test system because its honeycomb lattice, two-sublattice structure, edge orientation, and Dirac-like low-energy behavior produce rich transport characteristics while remaining suitable for tight-binding modeling.

2 Objectives

The project aimed to:

1. Review mesoscopic conductance from the Landauer viewpoint.
2. Relate scattering matrices and transmission probabilities to Green's functions.
3. Derive the recursive Green's function construction for a sliced device.
4. Explain how semi-infinite leads are represented through surface Green's functions and self-energies.
5. Apply the method to ballistic single-layer graphene.
6. Discuss the effect of graphene edge orientation and slicing efficiency on transport calculations.

3 Transport background

3.1 Ballistic conductance

For a ballistic conductor connected to contacts, conductance is determined by the number and transmission probability of available modes. In the simplest two-terminal case, the Landauer expression is

$$G = \frac{2e^2}{h} \sum_n T_n,$$

where T_n is the transmission probability of mode n . The factor of two accounts for spin degeneracy in the absence of spin splitting.

3.2 Landauer–Büttiker framework

For multiple terminals, the current in a terminal depends on transmission probabilities between all pairs of terminals. This formulation treats current and voltage probes within a common scattering framework and is useful for mesoscopic devices where probes can influence the local electronic distribution.

3.3 Green's functions and scattering

The retarded Green's function is written formally as

$$G^R(E) = [(E + i\eta)I - H]^{-1},$$

with η representing an infinitesimal or numerical broadening. In an open device, coupling to leads modifies the effective Hamiltonian through self-energies:

$$G^R(E) = [EI - H_C - \Sigma_L^R(E) - \Sigma_R^R(E)]^{-1}.$$

The level-width matrices are

$$\Gamma_{L,R} = i(\Sigma_{L,R}^R - \Sigma_{L,R}^A).$$

The transmission can then be expressed through a Fisher–Lee/Caroli-type relation involving G^R , G^A , and the lead broadenings.

4 Recursive Green's function method

4.1 Slicing the device

The finite conductor is divided into N slices. Each slice contains a finite number of sites and is described by an intra-slice Hamiltonian block h_n . Neighboring slices are coupled by matrices $U_{n,n+1}$ and $U_{n+1,n}$.

The slicing is chosen so that only neighboring slices interact. This block-tridiagonal structure is the basis of the recursion.

4.2 Left-connected recursion

Starting from the left lead, the method recursively constructs Green's functions for systems consisting of the left lead and the first n device slices. These quantities include the diagonal block of the left-connected Green's function and the blocks connecting the left boundary to the current slice.

4.3 Right-connected recursion

A corresponding right-to-left sweep begins from the right lead and constructs the Green's functions of systems connected to the right boundary.

4.4 Full Green's function

The left- and right-connected quantities are combined at each slice to obtain diagonal blocks of the full device Green's function. Boundary-to-boundary blocks are sufficient for transmission, while diagonal blocks are needed for local density of states.

4.5 Lead self-energies

Semi-infinite leads are incorporated through their surface Green's functions. The self-energy due to a lead has the general form

$$\Sigma = U^\dagger g^R U,$$

where g^R is the surface Green's function of the uncoupled semi-infinite lead and U is the coupling between the lead and the device.

The report discusses analytic methods, eigenmode decomposition, and iterative decimation for obtaining surface Green's functions.

5 Graphene model

5.1 Honeycomb lattice

Graphene is modeled as a single layer of carbon atoms arranged in a honeycomb lattice with two inequivalent sublattices, conventionally denoted A and B . Each site is connected to nearest neighbors on the opposite sublattice.

5.2 Tight-binding Hamiltonian

The nearest-neighbor model is written as

$$H = \sum_i V_i |i\rangle\langle i| + \sum_{\langle i,j \rangle} t_{ij} |i\rangle\langle j|,$$

where V_i is an onsite potential and t_{ij} is the hopping parameter between nearest-neighbor sites.

5.3 Edge orientations and slicing

The project considers armchair and zigzag edge geometries. Direct slicing can lead to many sites per slice and multiple inter-slice couplings. Alternative bent or optimized slicing schemes preserve the lattice connectivity while reducing the computational cost.

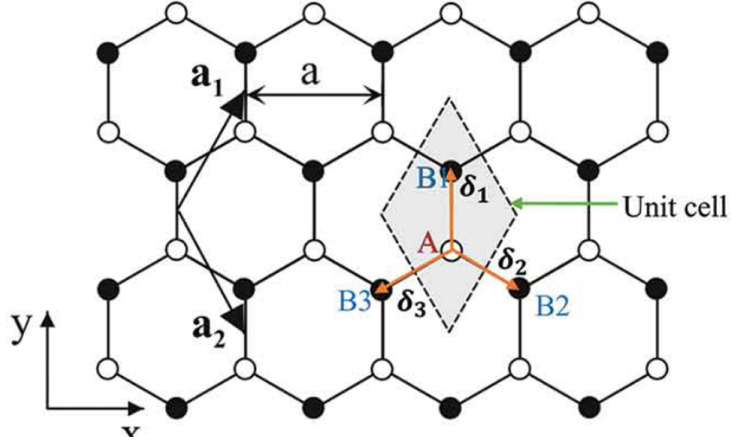


Figure 1: *Honeycomb graphene lattice with A/B sublattices.*

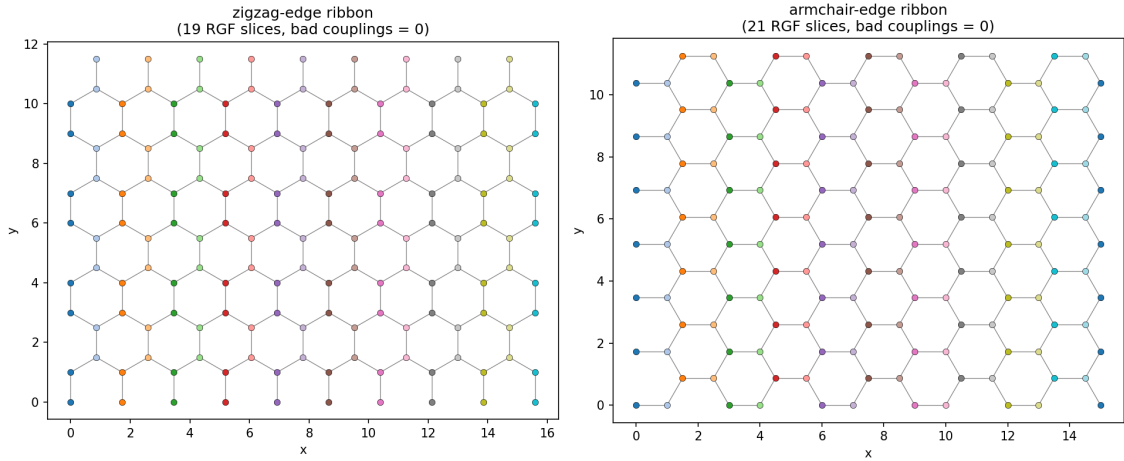


Figure 2: *Efficient slicing schemes for different graphene ribbons used to construct an efficient block-tridiagonal Hamiltonian: (a) armchair edge config. and (b) zigzag edge config. Each color denotes a single vertical layer.*

6 Observables

6.1 Transmission and conductance

Once the boundary Green's-function blocks are known, the transmission probability is obtained from the lead broadening matrices and the retarded/advanced Green's functions. The conductance follows from the Landauer formula.

6.2 Spectral function

The spectral function is

$$A(E) = i \left[G^R(E) - G^A(E) \right].$$

Its diagonal blocks provide information about the available local electronic states.

6.3 Local density of states

The local density of states at site n is obtained from the diagonal Green's-function element:

$$\rho_n(E) = -\frac{1}{\pi} \text{Im } G_{n,n}^R(E),$$

up to the sign and normalization convention used in the implementation.

7 Numerical results

7.1 Conductance of a clean armchair sample

The report calculates conductance as a function of the Fermi energy for a clean armchair graphene sample. The representative geometry has a width parameter corresponding to $M = 360$ and a length parameter corresponding to $N = 70$.

Furthermore, the conductance is also calculated for different (square lattice and honeycomb lattice) contacts with varying system sizes.

The results show good agreement with expected behavior for sufficiently large systems. Smaller devices exhibit stronger finite-size effects, including asymmetry and less sharply defined oscillatory structure.

7.2 Local density of states

The diagonal blocks of the full Green's function can be used to calculate the local density of states across the device. This provides spatial information that is not available from a conductance curve alone.

8 Interpretation

The project connects microscopic lattice structure with mesoscopic observables through a sequence of well-defined steps:

Lattice geometry $\rightarrow$ tight-binding Hamiltonian $\rightarrow$ sliced device blocks $\rightarrow$ lead self-energies
 $\rightarrow G^R \rightarrow$ transmission, conductance, and LDOS.

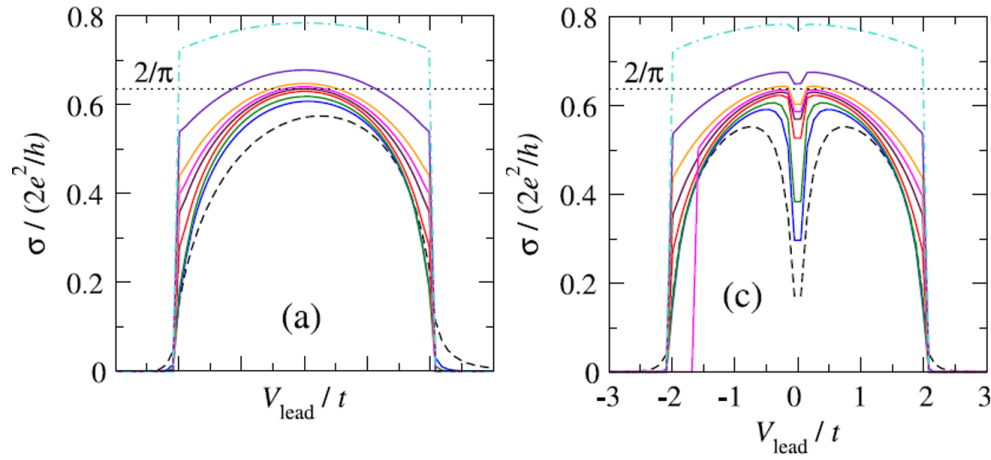
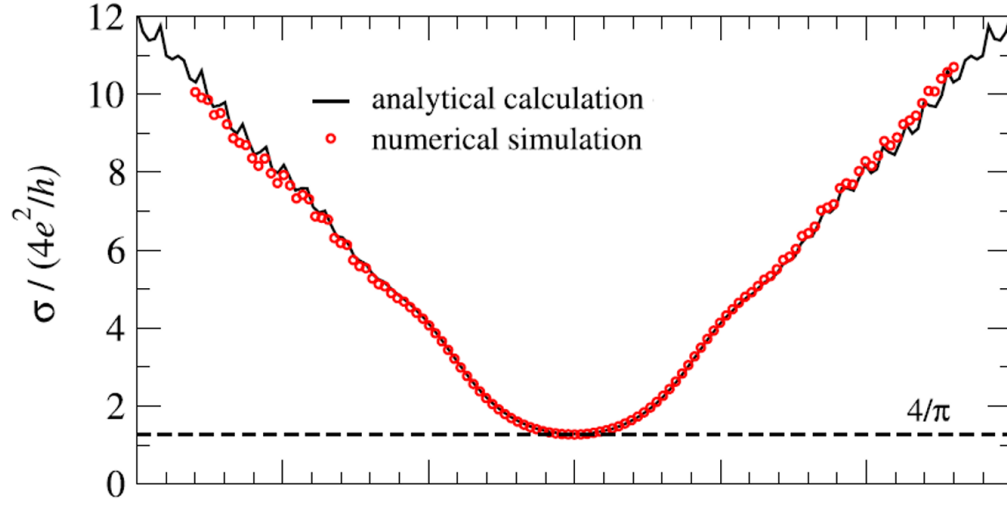


Figure 3: *Conductance of a graphene sample in the ballistic regime as a function of contact Fermi energy with $M = 360$: (a) clean armchair config. ($N = 70$) and (b) $N = 12$ (dashed line), 24, 36, 72, 96, 108, 120, 148, 200 for square lattice and honeycomb lattice contact respectively*

The principal computational advantage of RGF is that it avoids direct inversion of the full device matrix. Its computational cost is governed by the number of slices, the number of sites per slice, and the density of inter-slice couplings, making the choice of slicing a central aspect of the modeling strategy. Since slicing determines the dimension of the matrix blocks, it strongly influences the practical scalability of the calculation and makes RGF particularly suitable for larger quantum-transport simulations in which full-matrix methods become computationally expensive.

9 Scope and limitations

- The numerical demonstration focuses on clean, ballistic, single-layer graphene.
- The main tight-binding model includes nearest-neighbor hopping.
- Disorder, interactions, inelastic scattering, and finite-temperature effects are discussed but are not necessarily implemented in the numerical repository.
- The accuracy of conductance and LDOS depends on lead modeling, broadening, energy resolution, and boundary conditions.
- Finite-size effects become important for narrow or short devices.

Possible extensions include edge disorder, next-nearest-neighbor hopping, magnetic fields, multilayer graphene, finite-bias transport, self-consistent electrostatics, and comparison between direct inversion and RGF scaling.

10 Conclusion

This project develops a self-contained understanding of the recursive Green’s function method and demonstrates its application to ballistic graphene transport. The method provides an efficient route to conductance and local density of states by exploiting the block-tridiagonal structure of a sliced device and incorporating semi-infinite leads through self-energies.

The graphene application highlights the importance of lattice geometry, edge orientation, and slicing efficiency. The project therefore combines quantum transport theory, tight-binding modeling, Green’s-function methods, numerical linear algebra, and materials-focused computation.

References

- [1] C. H. Lewenkopf and E. R. Mucciolo, *The recursive Green’s function method for graphene*, Journal of Computational Electronics **12**, 203–231 (2013).
- [2] S. Datta, *Electronic Transport in Mesoscopic Systems*, Cambridge University Press (1997).
- [3] M. M. Odashima, B. G. Prado, and E. Vernek, *Pedagogical introduction to equilibrium Green’s functions: condensed-matter examples with numerical implementations*, Revista Brasileira de Ensino de Física **39** (2016).
- [4] D. S. Fisher and P. A. Lee, *Relation between conductivity and transmission matrix*, Physical Review B **23**(12), 6851 (1981).
- [5] H. Schomerus, E. Corenzwit, and L. D. Longinotti, *Effective contact model for transport through weakly-doped graphene*, Physical Review B – Condensed Matter and Materials Physics **76**(4), 045433 (2007).
- [6] J. Tworzydło, B. Trauzettel, M. Titov, A. Rycerz, and C. W. J. Beenakker, *Sub-Poissonian shot noise in graphene*, Physical Review Letters **96**(24), 246802 (2006).